WESTMINSTER
International University in Tashkent

System Analysis & Design

Lecture 2

Agenda

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1. Last week recap
2. Waterfall methodology

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Recap

What did we discuss last week?

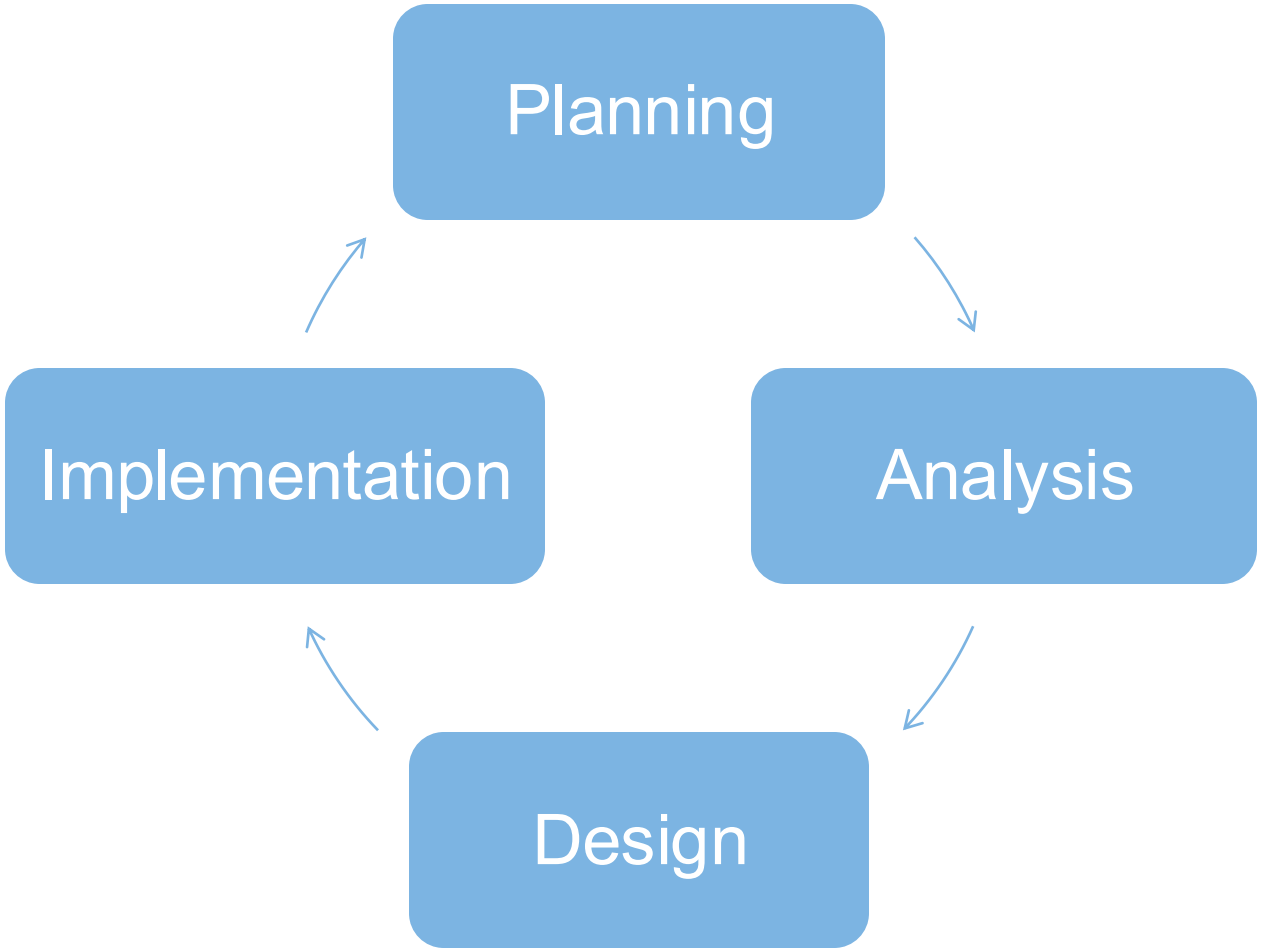
SDLC

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1. What does SDLC stand for?
2. What are the steps in SDLC?
3. What are the 3 major categories of Software development methodologies?

SDLC

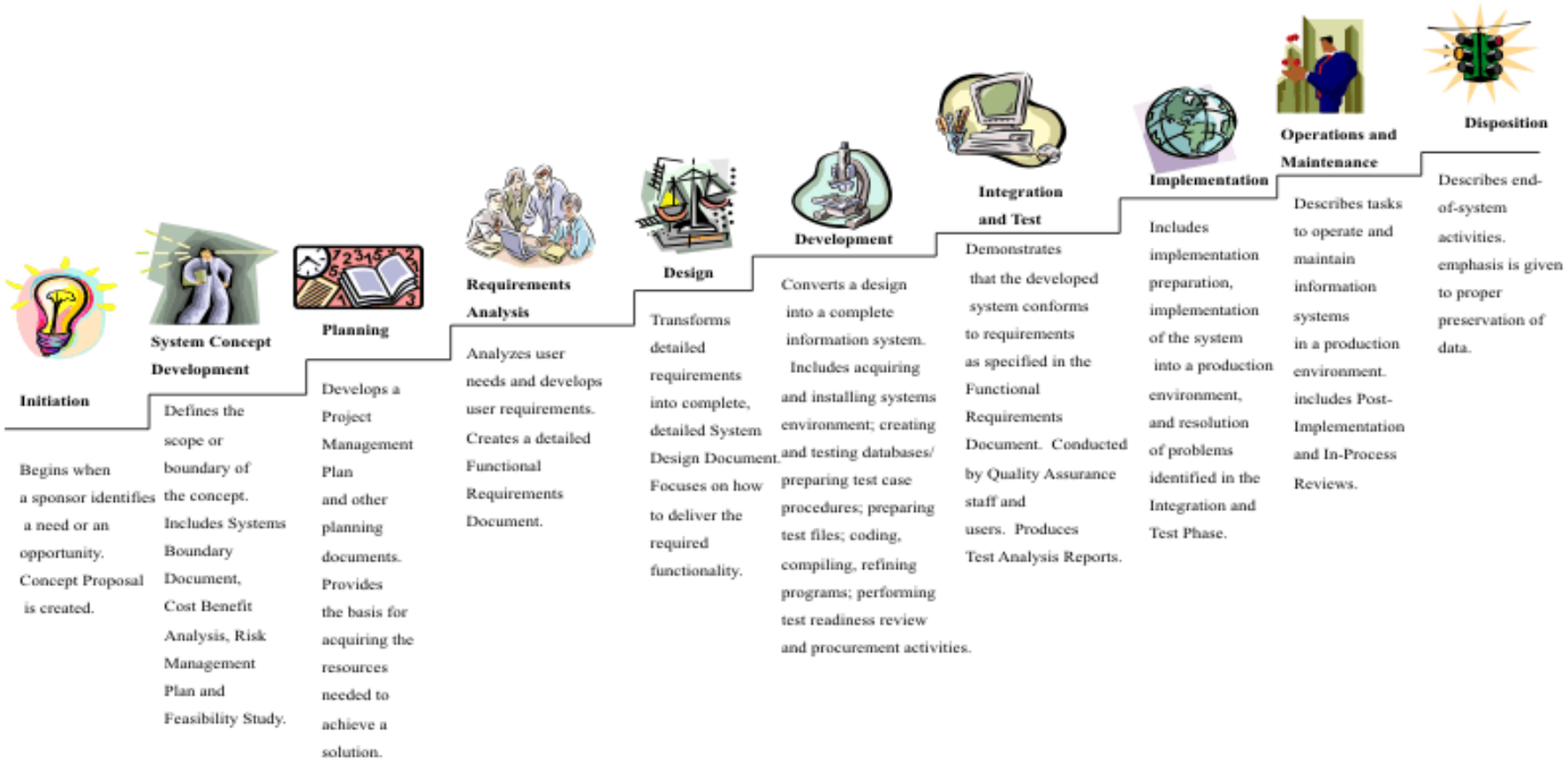
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SDLC

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Systems Development Life Cycle (SDLC)
Life-Cycle Phases



Categories

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There are many Software Development Models with their strengths and weaknesses. But they can be categorized into 3 major categories:

1. **Waterfall** – Structured, top/down approach.
   Examples: Waterfall, Structured Software Development Methodology (SSDM).

2. **Iterative & Incremental** – Iteratively go through important software development "steps" (analysis, design, build, etc.) and incrementally build the system.
   Examples: Spiral, RAD, Agile, etc.

3. **Prototyping** – Build prototypes, get feedback, refine. Once get the final version – build the software and deploy.
   Examples: RAD, Prototyping.

System Develop Methodologies

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Structural:

- Waterfall
- Structured Software Development Methodology (SSDM)

Iterative and Incremental:

- RAD
- Spiral
- eXtreme Programming
- Scrum
- Etc.

Prototyping:

- Prototyping

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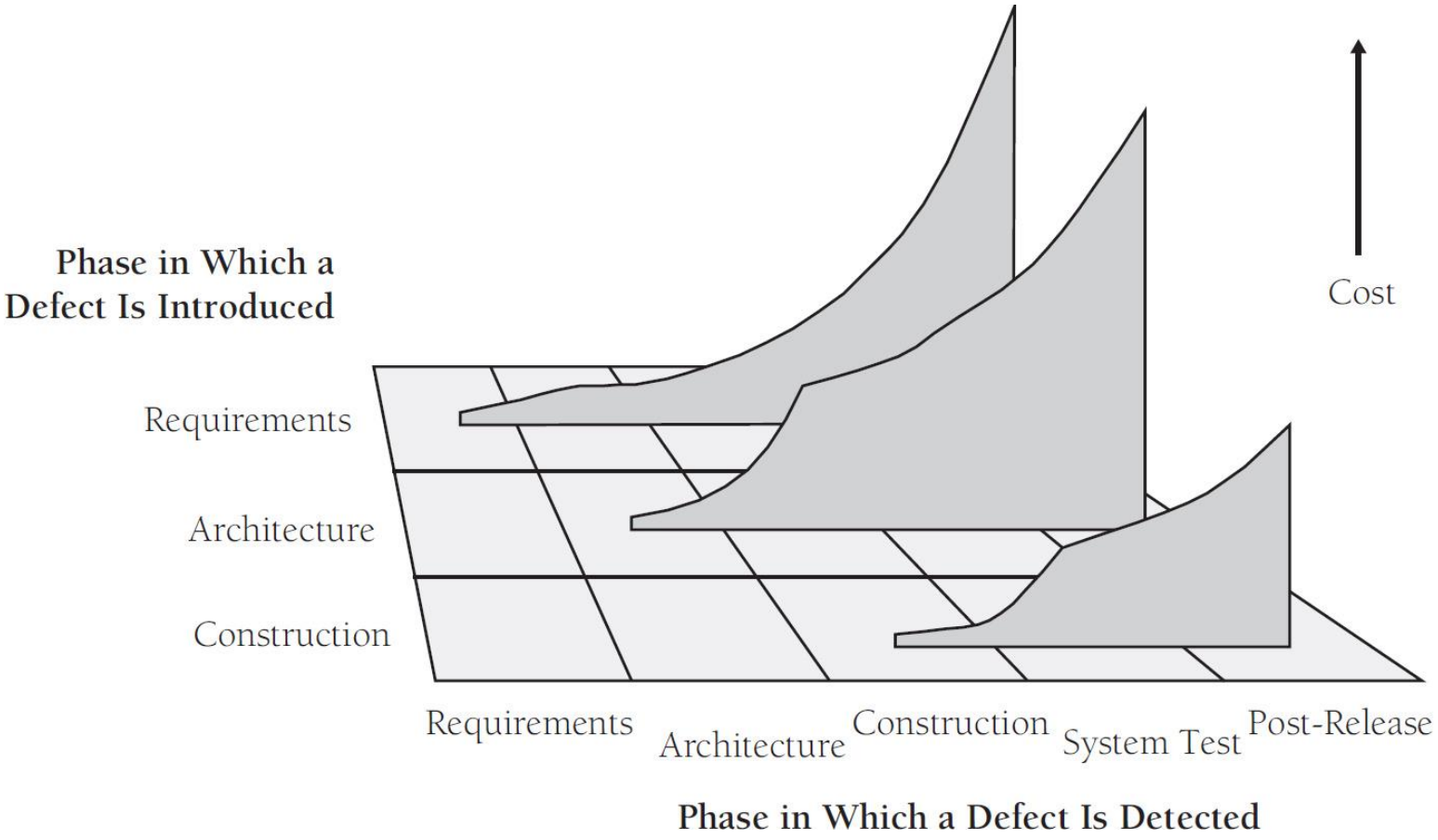
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Waterfall Model

Our first SDLC methodology

Cost of a “bug”

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Cost of a “bug”

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Time Detected					
Time Introduced	**Requirements**	**Architecture**	**Construction**	**System Test**	**Post-Release**
Requirements	1	3	5 – 10	10	10 – 100
Architecture	-	1	10	15	25 – 100
Construction	-	-	1	10	10 – 25

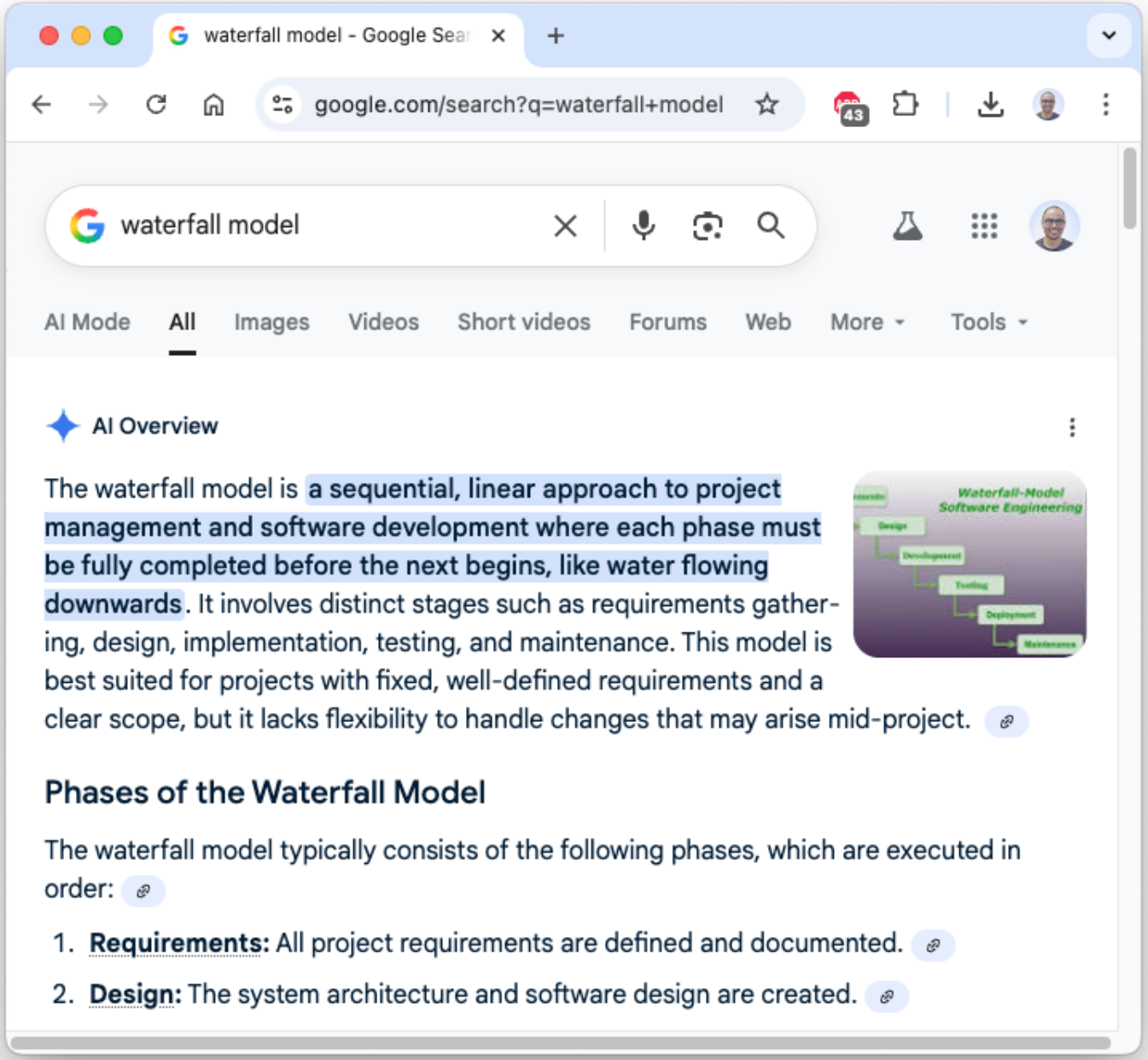
Source: Code Complete (Steve McConnell), Adapted from “Design and Code Inspections to Reduce Errors in Program Development” (Fagan 1976)

Waterfall Model

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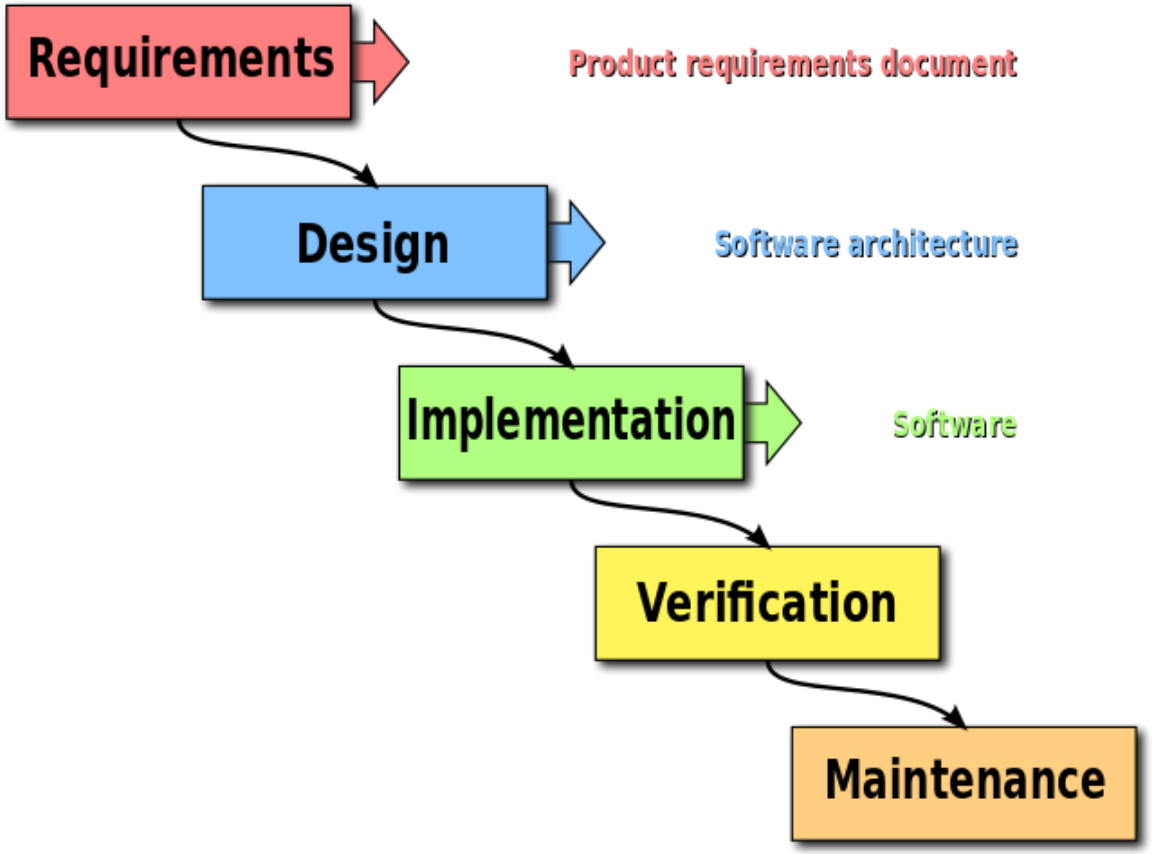
The waterfall model is a sequential design process, used in software development processes, in which progress is seen as flowing steadily downwards (like a waterfall) through the phases of conception, initiation, analysis, design, construction, testing, production/implementation and maintenance.

The waterfall development model originates in the manufacturing and construction industries: highly structured physical environments in which after-the-fact changes are prohibitively costly, if not impossible. Since no formal software development methodologies existed at the time, this hardware-oriented model was simply adapted for software development.


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AI Overview
The waterfall model is a sequential, linear approach to project management and software development where each phase must be fully completed before the next begins, like water flowing downwards. It involves distinct stages such as requirements gathering, design, implementation, testing, and maintenance. This model is best suited for projects with fixed, well-defined requirements and a clear scope, but it lacks flexibility to handle changes that may arise mid-project.
Waterfall-Model Software Engineering
Phases of the Waterfall Model
The waterfall model typically consists of the following phases, which are executed in order:
1. Requirements: All project requirements are defined and documented.
2. Design: The system architecture and software design are created.

Visual representation

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"Waterfall model" by Peter Kemp / Paul Smith - Adapted from Paul Smith's work at wikipedia.
Licensed under CC BY 3.0 via Commons - https://commons.wikimedia.org/wiki/File:Waterfall_model.svg#/media/File:Waterfall_model.svg

Model Steps

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1. **Requirements** – Documents all requirements.
2. **Design** – Based on the design, build a detailed system design.
3. **Implementation** – Build the system based on the design.
4. **Verification** – Test the new system and fix all the bugs.
5. **Maintenance** – Deploy and maintain the system.

Methodology Features

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- Steps are carried out in fixed order (sequencial).
- Each step must be fully finished before moving to the next (structured).
- Input of a step is the output of the previous step.
- Each step is documented. The output of each step is a documented.

Problems it solves

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Each step has an input and output.

A step's input is an output of the previous step.

What problems does it address?

- Changing project later on is costly. Tens or hundreds of times costly. That's why we need to spend time and resources to get them right at very first step.
- People may leave your team. They may be the carriers of the project related knowledge. That's why you need to document everything in a standard way.
- Gives concrete steps, that are easy to understand. Everyone knows at what phase the project is, and is the main goal at the moment – synergy.

When to use it?

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- Projects that have clearly defined final product.
- Stable and clear requirements i.e. requirements do not change.
- Clearly defined and fixed scope.
- Large scale project. For small projects it is not as effective.

What else do you think?

Advantages & Disadvantages

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From what you have heard so far…

What are the advantages of Waterfall methodology?

We will introduce **disadvantages** of the model, while introducing other methodologies. Usually, a new methodology is trying to solve specific problem in or that was not addressed in some methodology.



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Structured Software Development Methodology (SSDM)

Waterfall for Information Systems development.

System Develop Methodologies

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Iterative and Incremental:

- RAD
- Spiral
- eXtreme Programming
- Scrum
- Etc.

Prototyping:

- Prototyping

SSDM

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- SSDM is a Waterfall methodology for Information Systems (IS).
- It has the same “features” and follows the same ideas of Waterfall methodology:
  - Structured
  - Step-by-step
  - Heavily promotes documentation
  - Upfront planning and design
  - Etc.
- SSDM was rebranded as “Business System Development” in 2000.

Model Steps

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1. Feasibility
2. Investigation of the current environment
3. Business systems options
4. Definition of requirements
5. Technical system options
6. Logical design
7. Physical design

Stage 1: Feasibility

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- To understand if a given project is feasible, you need to do a feasibility analysis.

- There can be different feasibility analysis:
  - **No feasibility** – for small projects, where the environment is well understood and no need for feasibility analysis is required.
  - **Informal** – when the project doesn't have enough time or resources to spend, and it does not require documenting feasibility process.
  - **Formal** – you talk to users, identify business needs, problems, requirements and document it all in feasibility analysis report. This usually requires you to do initial requirements gathering.

Stage 1: Feasibility

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Concentrate on these areas:

- **Technical** – Is this project technically possible?
- **Financial** – Can the business effort it? Hardware, software, licenses, salaries for new staff, etc.
- **Organizational** – Will the organization adopt the new system? Is an organization willing to change/reorganize?
- **Ethical** – Will the final product raise ethical issues?

Also, don't forget to document the unfeasible, rejected solutions or versions of the project.

Stage 2: Investigate the current env

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- One of the most important stages. This helps the system analyst to understand the current user needs and problems.

- There are different techniques that you may employ:
  - Current system analysis
  - Read documentation
  - Interview
  - Questionnaires
  - Surveys
  - etc. *(we will talk more about these techniques in a separate lecture)*

Stage 3: Business system options

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- Equipped with the understanding of the current system in Stage 2, System Analyst tries to generate different “options” of the new system.

- Possible options:
  - Don’t do anything, leave the old system, everything is great.
  - Leave the old system, change business/operational rules.
  - Improve the existing system.
  - …
  - As the last resort – create a new system from scratch.

Stage 3: Business system options

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- You should try to get as many ideas and options as you can.

- Compile several “options” taking into the consideration all the information you gathered in previous stages.
  - Can the business afford this option?
  - Will it suite the organization?
  - Will users use it?
  - What changes must be made in organization?
  - etc.

- The product of this stage is (a single) chosen option.

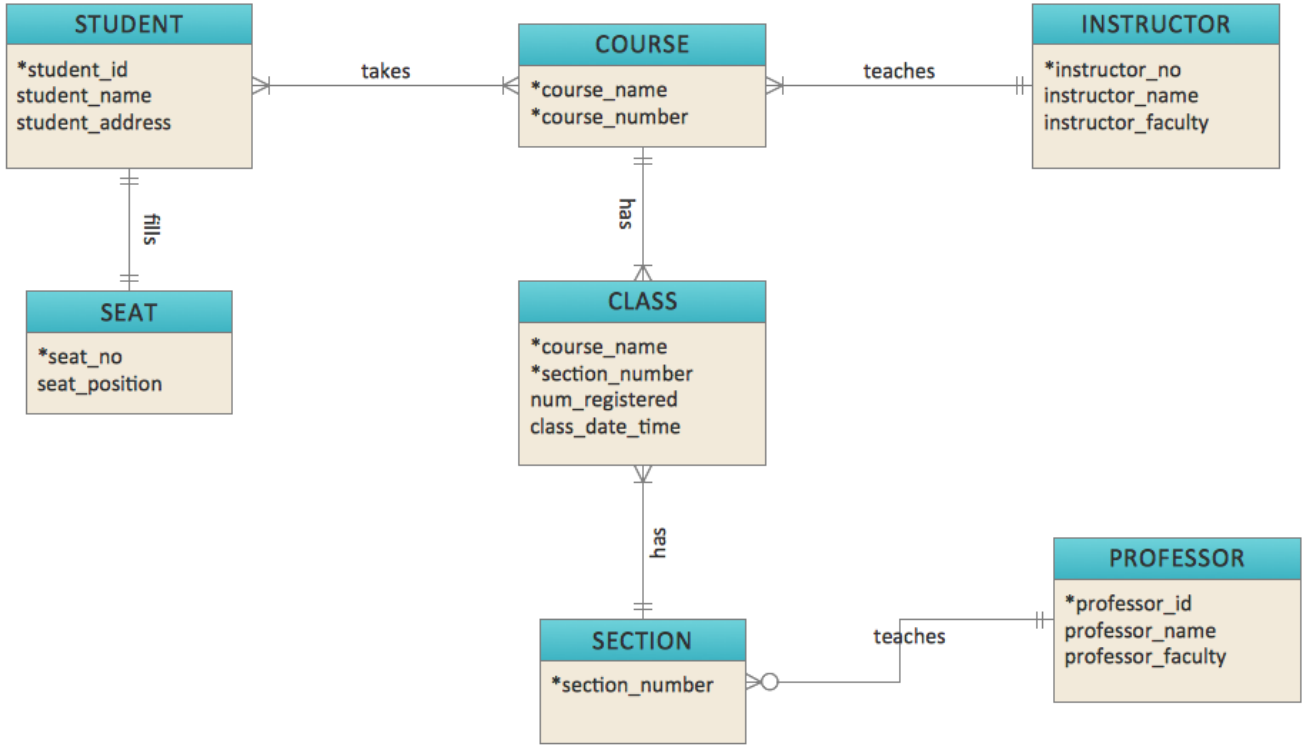
Stage 4: Definition of requirements WIUT

- Defines concrete, unambiguous and concise system requirements.

- At this stage you should produce:
  a) Logical Data Model – what the system will do, not how.
  b) Data Flow Diagram – what data is essential and its' relations.
  c) Entity Event Model – what are system actions and how they change the data in it?

Stage 4: Definition of requirements WIUT

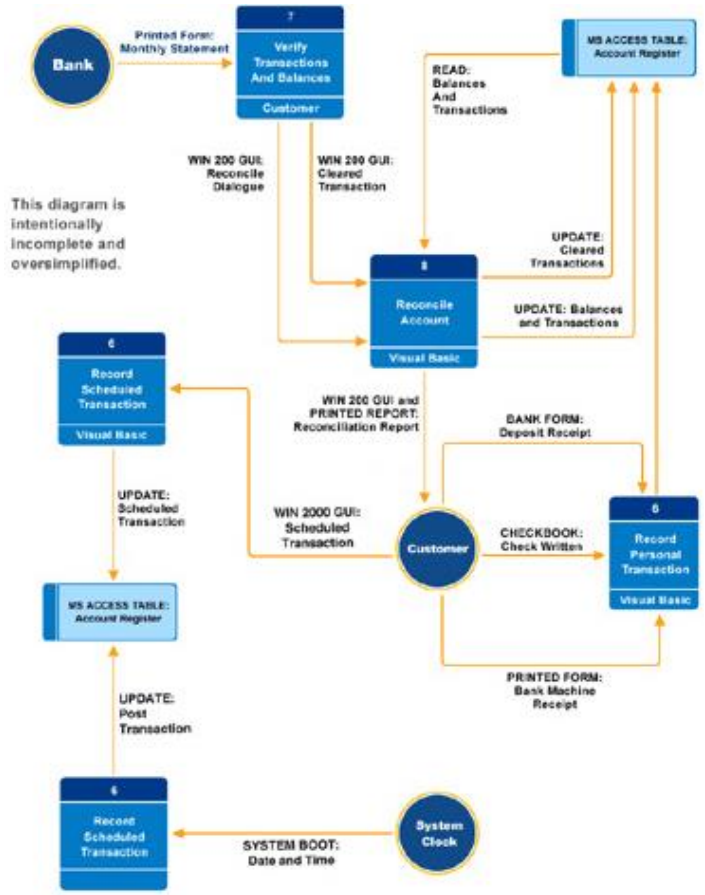
Logical Data Model Example:



Source: http://www.conceptdraw.com/solution-park/resource/images/solutions/entity-relationship-diagram-(erd)/Diagramming-Crow's-Foot-ERD-Sample60.png

Stage 4: Definition of requirements

Data Flow Diagram Example:



Source: http://www.jscgroup.com/images/ch-426-s.jpg

Stage 5: Technical system options

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- As in “Business system option” stage, you try to come up with several options to implement the new system. But, this time it is about technical aspects of the system.

- At the end, you would decide which “technical option” will be used.

Source: http://www.jscgroup.com/images/ch-426-s.jpg

Stage 5: Technical system options

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You might want to consider:

- What software will you use?
- What programming language?
- Hardware architecture
- Network architecture
- The cost of an option
- etc.

Source: http://www.jscgroup.com/images/ch-426-s.jpg

Stage 6: Logical Design

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- This step identifies and produces the specification on how users will interact with the system.
- In practice this means User Interface (UI).
- What techniques can you use at this stage?

Source: http://www.jscgroup.com/images/ch-426-s.jpg

Stage 7: Physical design

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- The final stage, where the system is physically implemented.
- Uses Logical Design stage outputs as an input. UI elements inherently contain business requirements defined in earlier stages.
- Output is the working system.

Source: http://www.jscgroup.com/images/ch-426-s.jpg

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Questions?


The End